



ARTIFICIAL INTELLIGENCE

screening models that will help in early risk detection.

NITI Aayog has also signed a statement of intent with various companies, including ABB, IBM and SAP, related to AI projects. It is also collaborating with Intel and Tata Institute of Fundamental Research (TIFR) on various AI-based research projects.

There are many Indian startups in the field of AI-based healthcare including Lybrate, mfine, Artelus, ChironX, LiveHealth, OnliDoc and Qure.ai.

- Lybrate is an online doctor consultation platform that connects patients with doctors and provides online consultation. Patients can also book lab tests and appointments online.

- mfine provides a virtual doctor with the ability to diagnose and triage common diseases. It refers patients with serious conditions to a hospital. Its goal is to provide healthcare to every Indian, including those in remote areas, through its virtual platform and AI software.

- OnliDoc uses AI and machine learning for end-to-end medical diagnosis. It has a phone application (available on Apple Store and Google Play Store) that can be used to find doctors, book appointments, and store medical records and prescriptions.

- LiveHealth is a management information system (MIS) for healthcare providers. It collects samples, manages patient records, diagnoses patients and generates reports.

Will AI replace doctors?

Over the course of various industrial revolutions, horses were replaced by machines yet there were no complaints. But think of machines replacing humans. Human doctors can make errors and many patients die due to such errors. Doctors suffer from fatigue, personal problems and other priorities at times. They are not machines that can do strenuous work and work long hours.

On the other hand, machines can do the job repeatedly without any complaint. These can detect skin, heart, eyes and other diseases. AI robots can do precision spinal cord surgery, too. AI is less invasive, faster and more accurate.

A recent report showed that an AI chatbot can diagnose medical conditions as accurately as a doctor. AI is also becoming more popular in areas such as radiology and cancer detection. All these new developments with AI raise the question: Will AI replace doctors and nurses in the future?

Co-founder of Sun Microsystems says that 80 per cent of doctors will be replaced by machines in the future. Some even think that bare hands may be banned and only AI robots will be allowed to do surgical procedures in the future.

AI replacing doctors in the near future is still a skeptical thought. A calculator can do amazing things at a fast speed and with accuracy, but that does not mean it can replace a mathematician. Doctors are required to develop medical AI.

Human emotion is the forte of human doctors. A company in Japan has developed emotional AI that can exhibit emotions. But patients need human touch, emotions and feelings. Human sympathy, empathy, perception and emotion are different from that of machines. So at this point of time, doctors are definitely going to stay!

What's next?

It is true that humans have a long record of failed quests for immortality. Can modern science and technology bring back the dead? Imagine a dead body kept in front of an AI machine. Now, let AI apply the most complex algorithm on the corpse. Lo and behold, the dead comes alive! Is that a possibility? No, that is creepy, and it can only be possible in imagination and in movies! AI cannot do wonders on the corpse, but it can revolutionise the medical industry in a big way.

Human organs and tissues can fail due to illnesses or accidents. Scientists can now repair damaged organs and tissues. There is a 3D bioprinting technique that prints human organs for replacing damaged organs. This means humans can now live longer.

AI has been created by humans—it is not a prophecy but a reality. And the reality is that we should expect the good, the bad and the ugly out of AI technology. There are speculations

that a day will come when AI robots will effectively take control of the planet away from the human species. Such imaginations can cause fear and trepidation in humans.

It is certain that medical advances and AI could potentially lead to massive benefits in healthcare and medical diagnoses. There are lots of problems in the healthcare industry, such as human errors and staff shortages. AI can fill the blind spots and eventually solve problems that we are facing today.

There are always pros and cons of any new and emerging technology. One disadvantage with AI could be a concern for privacy/security for patients because data could be on the cloud or connected devices. There could also be bias towards people who cannot afford Internet connectivity and access to high-tech data resources. Since AI is based on algorithms, anyone who has bad intentions could also develop an algorithm. Bad government policies could impede productivity and harm humans. Without proper regulatory bodies, implementation of AI technologies could also be at risk.

Conclusion

AI has a lot of potential, enormous benefits and opportunities, and is sure to bring many wonders to the healthcare space. Emergence of medical AI could radically change the course of human history and the healthcare industry. AI and 3D printing could replace damaged organs just like replacing parts in machines.

AI technology is making its presence felt in the healthcare industry at a very fast pace. India is one of the fastest adopters of AI-based healthcare services. One of the greatest potential benefits of AI is to help humans stay healthy so that a doctor is not required as often. AI will not fully replace human doctors but will certainly improve physician performance, and at the same time offer patients more accessibility to healthcare at lower costs. Proper regulations and a legal framework are required to make the best out of AI to serve mankind in a better and the most beneficial way. **EFY**